Zero Annotation Object Detection

Project Motivation

In my experience with object detection, I identified a major challenge: dependency on annotated datasets for training object detection models. Conventional approaches require extensive manual labeling of objects in images, which is time-consuming and costly. This becomes particularly problematic when dealing with new or unique datasets, such as medical samples or specialized objects, where annotated data isn't readily available.

The primary objective of this project is to enable the detection of objects or entities in images without relying on pre-existing annotations, thereby significantly reducing the time and effort typically required for data annotation. These heatmaps highlight regions of the image that the model considers important for making predictions, effectively allowing for the identification of objects or entities within the image, even in the absence of annotated training data.

The project begins by utilizing a pre-trained convolutional neural network (CNN) that has been fine-tuned for a specific image classification task which forms the basis for generating feature heatmaps from the convolution layers that highlight areas of interest within the images. To draw the bounding boxes, a few image processing techniques are performed, and the output is extracted from the generated feature heatmap.

The idea is that humans are only taught how to classify, and the ability to locate the entities comes inherently with birth, similarly, it is also possible to identify the location of the entity using only the classification model, but it also comes with a few image processing challenges. Questioning the existence of a similar ability of classification model gave rise to this project. A well-developed model can make entity detection and generate the annotations of the unannotated entities of an image.

Technical Implementation

In my implementation, I used a three-step approach to solve the zero-annotation challenge:

First, I employed a pre-trained MobileNetV2 model for initial feature extraction. Instead of relying on annotated bounding boxes, I used Gradient-weighted Class Activation Mapping (Grad-CAM) to understand which parts of the image the model finds important for classification.

Next, I developed a feature detection system that:

- Extracts activation maps from the model's convolutional layers

- Identifies regions of high activation as potential object locations

- Uses template matching to verify and refine detected regions

Finally, I implemented a bounding box generation algorithm that converts these detected regions into meaningful object boundaries without requiring any pre-defined annotations.

Dataset

This is a fruits dataset which has 9 fruit categories including apple, banana, cherry, chickoo, grapes, kiwi, mango, orange, and strawberry

Running Instructions

Required Libraries

- Python 3.12
- TensorFlow
- OpenCV
- PIL
- NumPy
- Matplotlib
- scikit-learn

Running the Code

1. Ensure the saved model file 'fruit_classifier.h5' after training is in the correct path
2. Update the image path in the notebook to your test image location
3. Run the notebook cells sequentially

Results

Through my experiments, I achieved several key outcomes:

Detection Performance:

- Successfully detected fruits in test images without using any annotated training data

- Generated accurate bounding boxes around detected objects
- System worked effectively across different fruit types

Feature Analysis:

- My implementation successfully extracted distinctive features from each fruit type
- The feature maps clearly showed important characteristics like shape and texture
- Grad-CAM visualizations confirmed the model's focus on relevant object parts

I tested the system on various fruit images and found it particularly effective at:

- Identifying object boundaries without manual annotations
- Handling different object orientations and positions
- Maintaining consistent detection quality across different fruit types

Conclusions

Based on my implementation results, I conclude that:

1. Zero-annotation object detection is feasible and can produce reliable results

2. The combination of deep learning features and traditional computer vision techniques provides a robust solution

3. This approach can significantly reduce the time and effort needed for implementing object detection systems

Moving forward, I recommend:

- Expanding the system to handle more complex object types
- Implementing real-time detection capabilities
- Optimizing the feature extraction process for better performance

Contributions

This project showcases a feature detection system built around a trained MobileNet fruit classifier. It provides a clear and practical approach to identifying and visualizing the key areas of an image that influence the model's decisions. By using Grad-CAM, it creates heatmaps that reveal what the model "sees," and combines this with preprocessing, bounding box drawing, and template matching to pinpoint features without needing annotations. The result is a versatile workflow that could be applied to real-world tasks like object detection in agriculture, retail, or even autonomous systems, making machine learning models more interpretable and useful.

Future scope

There is still huge room for improvement and make this more robust and perform well with advanced ML and image Processing techniques. When the whole code is in a stage of 100%, then it can be optimized to create annotations which are just images of different classes.